

[CubeSat System]

|— [OBC Subsystem : ESP32 + FreeRTOS]

| |— I²C Bus (MPU6050, BMP280)

| |— UART (NEO-6M)

| |— SPI (microSD)

| └─ Wi-Fi (Telemetry)

|— [Power Subsystem]

| |— Battery + BMS

| |— 3.3 V Regulator

| └─ Power Switch

|— [Communication Subsystem]

| └─ Web-based GCS (Laptop)

└─ [Structure Subsystem]

 └─ 1U Frame (BIRDS 4 compatible)